

DSAA3073: Theories in Data Science

Week 6 Open Investigation

Selecting a Clinical Risk Model Honestly

Information criteria under misspecification and adaptive reuse

Basic analysis. A 30-day readmission model for discharge planning

Use only the validation, AIC, BIC, and honest-evaluation distinctions from the Week 6 Learning Sheet

A hospital is developing a model to prioritize follow-up calls after discharge. The declared operational target is the conditional log score for a fresh patient from the same service population. Three regular logistic candidates were specified before fitting on $n = 400$ historical admissions. Their maximized log likelihoods and fitted parameter counts are

$$(\ell_1, q_1) = (-210, 3), \quad (\ell_2, q_2) = (-205, 6), \quad (\ell_3, q_3) = (-203, 10).$$

Candidate M_1 uses demographics and prior utilization; M_2 adds a small set of laboratory summaries; M_3 adds interaction terms. The team also holds a sealed prospective cohort that has not influenced preprocessing, the candidate list, or the reported likelihoods.

For the Basic analysis, provisionally assume independent patient records, regular identifiable maximum-likelihood fits, correct parameter counting, and a candidate regime in which the standard large-sample derivations are appropriate.

Basic Problem 1. Calculate the model-selection criteria and identify their targets

Compute AIC and BIC for all three models, using $\log(400) \approx 5.991$. State the preferred candidate under each criterion. Explain why a disagreement is not an arithmetic contradiction: identify the inferential target associated with each rule.

Basic analysis. Separate selection from final evidence

The selected criterion value does not become an untouched performance estimate

Basic Problem 2. Recommend a complete analysis rather than a criterion

The hospital's stated goal is predictive conditional log score, not identification of a true finite-dimensional model. Choose a provisional candidate and describe the remaining data flow. State precisely what may be reported from the development sample and what must be obtained from the sealed cohort. Include every adaptive choice in your definition of the procedure being evaluated.

Thinking investigation. Formalize target, rule, and guarantee

A selection formula is valid only relative to a sampling model and a scientific target

Write a model-selection analysis as

$$(\mathcal{M}, \mathcal{T}, \mathcal{A}, \mathcal{S}, \mathcal{G}),$$

where $\mathcal{M}$ is the candidate family, $\mathcal{T}$ the target, $\mathcal{A}$ the asymptotic or sampling assumptions, $\mathcal{S}$ the selection rule, and $\mathcal{G}$ the supported conclusion. The Basic AIC calculation uses regular maximum likelihood and the information equality.

Investigation 1. Allow likelihood misspecification

Define the pseudo-true parameter θ^* as a maximizer of the population expected log likelihood, equivalently a minimizer of the candidate's Kullback-Leibler discrepancy from the data-generating conditional distribution. Let $s_{i(\theta)} = \nabla \log f(Y_i \text{ mid } X_i, \theta)$, and define

$$A = \mathbb{E}[-\nabla^2 \log f(Y \text{ mid } X, \theta^*)], \quad B = \mathbb{E}[s(\theta^*)s(\theta^*)^T].$$

Assume these matrices are finite and A is nonsingular. Standard AIC uses the equality $A = B$. Suppose the candidate is misspecified, so this equality need not hold. Does the $2q$ optimism correction remain justified? Derive the matrix expression that replaces it and show how the standard formula is recovered under information equality.

Thinking investigation. Remove the true-model-in-list assumption

The identification claim fails, but an asymptotic comparison may still be derived

Investigation 2. Determine what BIC selects when every candidate is wrong

Suppose the candidate set is finite and regular, but the data-generating distribution is not represented by any candidate. Let K_j be the minimum Kullback-Leibler discrepancy of candidate M_j from the data-generating distribution. Derive the leading pairwise difference $\text{BIC}_j - \text{BIC}_k$. You may denote by $o_{p(n)}$ a random remainder whose ratio to n converges in probability to zero. If one candidate has strictly smallest K_j , what dominates as n grows? What part of the usual BIC conclusion is no longer available?

Thinking investigation. Reuse selection scores as evaluation

Unbiasedness for fixed candidates is destroyed by taking the observed minimum

Investigation 3. Derive exact selection optimism in a two-candidate model

Consider two procedures with the same true risk R . Their validation estimates are

$$\hat{R}_1 = R + \tau Z_1, \quad \hat{R}_2 = R + \tau Z_2,$$

where Z_1, Z_2 are independent standard normal variables. Each validation estimate is unbiased before selection. The analyst chooses the smaller score and reports that same minimum as final performance. Derive $\mathbb{E}[\min(\hat{R}_1, \hat{R}_2)]$ exactly. Explain which Week 6 statement remains true and which one fails after selection.

Synthesis. State the general Week 6 conclusion

For the hospital case, distinguish three failures: likelihood misspecification, absence of a true candidate, and adaptive reuse of evaluation scores. For each, state the strongest analytic object that remains available and one empirical diagnostic or design change that addresses the failure.